O-D-1833
March 1, 1976

FEDERAL SPECIFICATION

DISINFECTANT-DETERGENT, GERMICIDAL AND FUNGICIDAL CONCENTRATE (IODOPHOR-TYPE)

This specification was approved by the Commissioner, Federal Supply Service, General Services Administration, for the use of all Federal agencies.

1. SCOPE

1.1 Scope. This specification covers one grade of disinfectant-detergent concentrate which contains iodine as the active ingredient (see 6.1).

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issues in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

Federal Specifications:

GG-H-80 - Handle and Blade Set, Surgical Knife, Detachable Blade.
SS-T-312 - Tile, Floor: Asphalt, Rubber, Vinyl, Vinyl-Asbestos.
LLL-F-1238 - Floor Covering, Linoleum.
PPP-B-636 - Boxes, Shipping, Fiberboard.

Federal Standard:

Fed. Std. No. 123 - Marking for Shipment (Civil Agencies).

(Activities outside the Federal Government may obtain copies of Federal Specifications, Standards, and Handbooks as outlined under General Information in the Index of Federal Specifications and Standards and at the prices indicated in the Index. The Index, which includes cumulative monthly supplements as issued, is for sale on a subscription basis by the Superintendent of Documents, U.S. Government Printing Office, Washington, DC 20402.

(Single copies of this specification and other Federal Specifications required by activities outside the Federal Government for bidding purposes are available without charge from Business Service Centers at the General Services Administration Regional Offices in Boston, New York, Washington, DC, Atlanta, Chicago, Kansas City, MO, Fort Worth, Denver, San Francisco, Los Angeles, and Seattle, WA.

(Federal Government activities may obtain copies of Federal Specifications, Standards, and Handbooks and the Index of Federal Specifications and Standards from established distribution points in their agencies.)

Military Standards:

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129 - Marking for Shipment and Storage.

(Copies of Military Specifications and Standards required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

FSC 6840

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T11:16Z
Check the source to verify that this is the current version before use.

Federal Regulations:

16 CFR 1500 - Hazardous Substances and Articles: Administration and Enforcement Regulations.
40 CFR 162 - Regulations for the Enforcement of the Federal Insecticide, Fungicide, and Rodenticide Act.

(The Code of Federal Regulations (CFR) and the Federal Register (FR) are for sale on a subscription basis by the Superintendent of Documents, U.S. Government Printing Office, Washington, DC 20402. When indicated, reprints of certain regulations may be obtained from the Federal agency responsible for issuance thereof.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless a specific issue is identified, the issue in effect on date of invitation for bids or request for proposal shall apply.

Academic Press:

Methods of Animal Experimentation, Volume I, edited by William I. Gay (1965).

(Application for copies should be addressed to Academic Press, Inc., 111 Fifth Avenue, New York, NY 10003.)

American Society for Testing and Materials (ASTM) Standards:

D 1310 - Standard Method of Test for Flash Point of Liquids by Tag Open-Cup Apparatus.
E 70 - Standard Method of Test for pH of Aqueous Solutions with the Glass Electrode.

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA. 19103.)

Association of Official Analytical Chemists (AOAC):

Official Methods of Analysis.

(Application for copies should be addressed to the Association of Official Analytical Chemists, P.O. Box 540, Benjamin Franklin Station, Washington, DC 20044.)

National Fire Codes (NFC):

NFC No. 56A, Part IV - Standard for the Use of Inhalation Anesthetics: Requirements for Flammable Anesthetizing Locations and Flammable Agent Storage Locations.

(Application for copies should be addressed to the National Fire Protection Association, 60 Batterymarch Street, Boston, MA 02110.)

National Motor Freight Traffic Association, Inc., Agent:

National Motor Freight Classification.

(Application for copies should be addressed to the American Trucking Associations, Inc., Tariff Order Section, 1616 P Street, N.W., Washington, DC 20036.)

Uniform Classification Committee, Agent:

Uniform Freight Classification.

(Application for copies should be addressed to the Uniform Classification Committee, Room 1106, 222 South Riverside Plaza, Chicago, IL 60606.)

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T11:16Z
Check the source to verify that this is the current version before use.

3. REQUIREMENTS

3.1 Registration. The concentrate shall be registered in accordance with the applicable section of 40 CFR 162.

3.2 Material. The concentrate shall be a liquid.

3.3 Biological requirements.

3.3.1 Germicidal activity. The concentrate shall kill the following organisms when tested as specified in 4.3.2.1.

(a) Salmonella choleraesuis (ATCC No. 10708).
(b) Staphylococcus aureus (ATCC No. 6538).
(c) Pseudomonas aeruginosa (ATCC No. 15442).

3.3.2 Tuberculocidal activity. The concentrate shall kill Mycobacterium tuberculosis var. bovis (BCG) when tested as specified in 4.3.2.2.

3.3.3 Fungicidal activity. The concentrate shall kill Trichophyton mentagrophytes (ATCC No. 9533) when tested as specified in 4.3.2.3.

3.3.4 Acute oral toxicity. The concentrate shall kill not greater than five of a group of ten rats, when tested as specified in 4.3.2.4.

3.3.5 Dermatitis. The concentrate shall cause no primary irritation to the skin when tested as specified in 4.3.2.5.

3.4 Non-injurious requirements.

3.4.1 Conductive flooring. The concentrate shall not cause the resistance of conductive flooring to exceed the limits specified in the section on the reduction in electrostatic hazard of conductive flooring of NFC No. 56A, Part IV. Resistance shall be measured as specified in 4.3.3.1.

3.4.2 Tile. The concentrate shall cause no softening, swelling, cracking, or permanent discoloration of tile when tested as specified in 4.3.3.2.

3.4.3 Metal. The concentrate shall have no corrosive effect on stainless steel and not greater than 12 mg loss in weight of galvanized iron when tested as specified in 4.3.3.3 and 4.3.3.4, respectively.

3.5 Chemical and physical requirements.

3.5.1 Available iodine. The concentrate shall provide not less than 1.60 and not greater than 2.18 percent available iodine when tested as specified in 4.3.4.1.

3.5.2 Rinsability. The concentrate shall leave a residual film not greater than the hard water control when tested as specified in 4.3.4.2.

3.5.3 Flash point. The concentrate shall have flash point not lower than 64°C (148°F) when tested as specified in 4.3.4.3.

3.5.4 Color. The concentrate shall be a clear, definite amber color which shall provide visual indication of potency. It shall contain no colored inert ingredient when tested as specified in 4.3.4.1 (see footnote to 4.3.4.1).

3.5.5 Odor. The concentrate shall leave no lasting or objectionable odor when tested as specified in 4.3.4.4.

3.5.6 Storage stability. The concentrate shall show no separation and shall contain not less than 1.60 and not greater than 2.18 percent available iodine when tested as specified in 4.3.4.5.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T11:16Z
Check the source to verify that this is the current version before use.

3.5.7 pH. The concentrate shall have pH not less than 3 and not greater than 4 when tested as specified in 4.3.4.6.

3.5.8 Clarity. The concentrate shall show no evidence of precipitation when tested as specified in 4.3.4.7.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure that supplies and services conform to prescribed requirements.

4.2 Quality conformance inspection.

4.2.1 Examination of preparation for delivery.

4.2.1.1 Examination of packing.

4.2.1.1.1 Lot. For this examination, a lot shall be concentrate, fully packed, marked, and offered for examination at one time. Lot size shall be expressed in number of filled shipping containers.

4.2.1.1.2 Sampling plan. Sample unit for this examination shall be one filled shipping container. Sampling shall be conducted in accordance with MIL-STD-105. Sample size, acceptance and rejection numbers shall be determined by special inspection level S-2 and acceptable quality level (AQL) 4.0 percent defective, respectively. Defects shall be scored in accordance with table I.

TABLE I. Examination of packing

Examine	Defect
Marking	Missing, incorrect, illegible; not as specified.
Contents	Net contents less than specified.
Workmanship	Closure incomplete or inadequate; loose or inadequate strapping or stapling; bulged, distorted, or otherwise defective container; not as specified.

4.2.1.2 Examination of packaging.

4.2.1.2.1 Lot. For this examination, a lot shall be concentrate, fully packaged, marked, labeled, and offered for examination at one time. Lot size shall be expressed in number of filled unit packages.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T11:16Z
Check the source to verify that this is the current version before use.

4.2.1.2.2 Sampling plan. Sampling shall be as specified in 4.2.1.1.2, except that the sample unit shall be one unit package and defects shall be scored in accordance with table II.

TABLE II. Examination of packaging

Examine	Defect
Marking	Missing, incorrect, illegible; not as specified.
Labeling	No EPA registration number.
Contents	Net contents less than specified.
Workmanship	Closure incomplete or inadequate: leakage or seepage from any part of the container; bulged, distorted, or otherwise defective container; not as specified.

4.2.2 Examination of material and workmanship.

4.2.2.1 Lot. For this examination, a lot shall be concentrate produced by one manufacturer, at one plant, from the same raw materials, by the same manufacturing process, and offered for examination at one time. In the event of a batch process, each batch shall constitute a lot. Lot size shall be expressed in equivalent number of filled unit packages.

4.2.2.2 Sampling plan. Sample size for this examination shall be one unit package or equivalent volume drawn at random from the lot. Defects shall be determined in accordance with table III. Lot shall be rejected if sample contains one defect.

TABLE III. Examination of material and workmanship

Examine	Defect
Material	Not a liquid.
Workmanship	Foreign matter visibly present: not suitable for the intended use.

4.2.3 Testing.

4.2.3.1 Lot. A lot shall be as specified in 4.2.2.1.

4.2.3.2 Sampling plan. Sample size for testing shall be one unit package or an equal volume, selected at random from the lot. Sample shall be tested as specified herein. Lot shall be rejected if the sample fails to pass any test.

4.3 Test methods.

4.3.1 Testing conditions.

4.3.1.1 Synthetic hard water. When dilution is required, synthetic hard water equivalent to 100 ppm hardness, calculated as $CaCO_3$, shall be used. It shall be prepared in accordance with the AOAC method for the preparation of synthetic hard water for germicidal and detergent sanitizers.

4.3.1.2 Test dilutions. Except where otherwise specified (see 4.3.2.2, 4.3.2.4, 4.3.2.5, 4.3.4.1, 4.3.4.3, 4.3.4.5, 4.3.4.7), dilutions in the ratio 1:213::concentrate: synthetic hard water shall be used for testing.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T11:16Z
Check the source to verify that this is the current version before use.

4.3.2 Biological testing.

4.3.2.1 Germicidal activity. Germicidal activity for the organisms in 3.4.1 shall be determined in accordance with the AOAC use-dilution method for disinfectants.

4.3.2.2 Tuberculocidal activity. Tuberculocidal activity for the organism in 3.4.2 shall be determined in accordance with the AOAC method for the determination of tuberculocidal activity of disinfectants, confirmative in vitro test, except that plain distilled water may be used instead of the synthetic hard water for the test dilution.

4.3.2.3 Fungicidal activity. Fungicidal activity for the organism in 3.4.3 shall be determined in accordance with the AOAC fungicidal test for disinfectants.

4.3.2.4 Acute oral toxicity. Ten healthy male albino rats of the Sprague Dawley or Wistar strain, weighing between 175 to 225 gm, shall be used for the oral toxicity test. The animals shall be fasted for 24 hours preceding dosing. The test rats shall be weighed. A syringe, to which a metal feeding tube has been attached shall be filled with a measured amount of the undiluted concentrate (4 ml per kilogram of body weight). The rats shall then be dosed orally using the infusion technique for administration of fluids via the digestive tract per os for the rat, as described in Methods of Animal Experimentation. After administration of the concentrate, place the animals in their normal environment with usual food and water. Keep the animals under observation for a period of one week.

4.3.2.5 Dermatitis. Primary irritation to the skin shall be determined in accordance with the method of testing primary irritant substances in 16 CFR 1500.

4.3.3 Non-injurious testing.

4.3.3.1 Conductive flooring. The resistance of the conductive flooring shall be measured as specified in NFC 56A, Part IV. In lieu of the test, the supplier may submit evidence of listing with Underwriter's Laboratories.

4.3.3.2 Tile. Test specimens, approximately 2 x 4 x 1/8 inches (50 x 100 x 3mm), as specified below, shall be used:

White linoleum, conforming to LLL-F-1238, type I, class 2, grade A.
Black asphalt, conforming to SS-T-312, type I.
Black rubber, conforming to SS-T-312, type II.
Black vinyl-asbestos, conforming to SS-T-312, type IV.

Each specimen shall be immersed in a separate 250-ml beaker containing 150 ml of the test dilution. The specimens shall remain in the dilution for six hours at room temperature. At the end of this time, the specimens shall be removed, rinsed thoroughly with tap water, given a final rinse with distilled water, and dried at room temperature. They shall be visually examined for softening, swelling, cracking, or discoloration.

4.3.3.3 Stainless steel. A scalpel blade conforming to GG-H-80 shall be completely immersed in the test dilution for six hours at room temperature. At the end of this time, the blade shall be removed, thoroughly rinsed with tap water, given a final rinse with distilled water, and dried at room temperature. It shall be visually examined for signs of corrosion.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T11:16Z
Check the source to verify that this is the current version before use.

4.3.3.4 Galvanized iron. Six 2 x 2 inch (50 x 50 mm) test specimens shall be cut from 18 gauge galvanized iron sheet and washed in carbon tetrachloride. Immediately before the test, the test specimens shall be wiped with a clean cloth moistened with carbon tetrachloride, dried at 105°C for 30 minutes, and cooled in a dessicator. Each test specimen shall be accurately weighed to the nearest 0.1 milligram and completely immersed, two to a beaker, in three beakers containing the test dilution. The test specimens shall not touch each other while immersed. The beakers shall be covered with a watch glass and immersed to the solution level in a constant temperature water bath maintained at 35 + 1°C for six hours. At the end of this time, the specimens shall be removed from the dilutions, rinsed thoroughly with tap water, given a final rinse with distilled water, wiped with a clean towel, and dried at 105°C for 30 minutes. The specimens shall be cooled in a dessicator and re-weighed. The weight loss shall be calculated by:

$$\text{Average weight loss} = \sum_{i=1}^{6} \frac{(W_1 - W_2)_i}{6}$$

Where:

W_1 = weight before immersion.
W_2 = weight after immersion.

4.3.4 Chemical and physical testing.

4.3.4.1 Available iodine. Place approximately 20 grams, weighed to the nearest 0.01 g, of the undiluted concentrate in a clean, 250-ml wide-mouth erlenmeyer flask. Add 25 ml distilled water, mix thoroughly, then titrate the iodine solution with standard .1N- sodium thiosulfate until the solution just becomes colorless[1] and the amber color of the iodine does not reappear for one full minute. Calculate the percentage of available iodine by:

$$\text{Percent available iodine} = \frac{12.7\ NV}{W}$$

Where:

N = normality of sodium thiosulfate.
V = Volume, ml, of sodium thiosulfate.
W = sample weight, g.

1/ At this time, the presence of colored inert ingredients shall also be determined. Inability to obtain the colorless end point shall be interpreted as the presence of colored inert ingredients.

4.3.4.2 Rinsability. Approximately 100 ml of the test dilution shall be placed in a clean, dry, 250-ml erlenmeyer flask. The flask shall be stoppered, shaken vigorously for 10 seconds, and the dilution poured-out. The flask shall be thoroughly rinsed with synthetic hard water and allowed to dry. A second clean, dry, 250-ml erlenmeyer flask shall be thoroughly rinsed with synthetic hard water, allowed to dry, and compared with the first flask for residual film.

4.3.4.3 Flash point. The flash point of the undiluted concentrate shall be determined in accordance with ASTM D 1310.

4.3.4.4 Odor. A six-inch square of bleached cotton sheeting shall be completely immersed in the test dilution. After having been thoroughly wetted, the sheeting shall be removed, wrung to remove excess liquid, and hung to dry for 48 hours in a well ventilated room at the end of this time, examine for residual odor.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T11:16Z
Check the source to verify that this is the current version before use.

4.3.4.5 Storage stability. The undiluted concentrate shall be stored in a glass-stoppered bottle for two weeks at 50° + 2°C. At the end of this period it shall be allowed to come to room temperature and examined for phase separation and presence of precipitates. The concentrate shall contain not less than 1.60 and not greater than 2.18 percent available iodine when tested as specified in 4.3.4.1.

4.3.4.6 pH. The pH shall be determined in accordance with ASTM E 70.

4.3.4.7 Clarity. The undiluted concentrate shall be diluted with distilled water in the ratio 1:1280::concentrate:distilled water. The dilution shall be thoroughly mixed and stored in a clear, glass-stoppered vessel at room temperature for four hours. At the end of that time, the dilution shall be examined for presence of precipitates.

5. PREPARATION FOR DELIVERY

5.1 Packaging. Packaging shall be level A, B, or C, as specified (see 6.2).

5.1.1 Levels A and B. One (1) gallon of concentrate shall be furnished in a high density polyethylene (0.95 density minimum) container with handle and screwcap-type closure. The screwcap shall be secured with a cellulose hydrate band, heat-shrinkable polyvinyl chloride tubing, or acid-resistant vinyl chloride, self-adhering pressure-sensitive tape, which covers the bead of the container and extends at least halfway up the sides of the screwcap. This outer seal shall adhere both to the cap and to the neck of the container to prevent loosening of the cap. Alternatively the container may be sealed with an aluminum heat seal foil and screwcap combination without self-adhering tape. This seal should prevent leaking of concentrate even if the screwcap were to be removed.

5.1.2 Level C. Concentrate shall be packaged to afford adequate protection against damage during shipment from the supply source to the first receiving agency.

5.2 Packing. Packing shall be level A, B, or C, as specified (see 6.2).

5.2.1 Level A. Four (4) one-gallon containers shall be packed in a box conforming to PPP-B-636, class weather-resistant, grade V3c or W5c, as specified (see 6.2). The box shall be fitted with dividers or partitions to provide a cell for each container. The dividers or partitions shall be of the same class of material as the box and be the same height as the box. The box shall be not less than 3/8-in higher than the container. The box shall be closed, waterproofed, and banded in accordance with the appendix of PPP-B-636.

5.2.2 Level B. Four (4) one-gallon containers shall be packed in a box conforming to PPP-B-636, class domestic. The box shall be fitted with dividers or partitions as specified in 5.2.1, and shall be closed in accordance with the appendix of PPP-B-636.

5.2.3 Level C. Concentrate, packaged as specified in 5.1, shall be packed in containers to assure carrier acceptance and safe arrival at destination. Containers shall conform to National Motor Freight Classification or Uniform Freight Classification as applicable.

5.3 Marking. Marking of interior packages and exterior shipping containers shall be in accordance with Fed. Std. No. 123 or MIL-STD-129, as specified (see 6.2).

5.4 Labeling. Labeling shall be in accordance with the corresponding section of 40 CFR 162.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T11:16Z
Check the source to verify that this is the current version before use.

6. NOTES

6.1 Intended use. The concentrate covered by this specification is intended for use as a general disinfectant of environmental surfaces. It is not intended for use as a surgical scrub. It should be diluted with water according to the manufacturer's suggestion and may be used to clean and disinfect floors, washable walls, metal furniture, and elsewhere throughout cafeterias, locker rooms, and sick rooms. For better results, gross filth and heavy soil deposits should be cleaned from surfaces prior to disinfecting with the diluted solution of concentrate. Food processing and beverage equipment should be rinsed thoroughly with potable water after disinfecting.

6.2 Ordering data. Purchasers should select the preferred options permitted herein, and include the following information in procurement documents:

(a) Title, number, and date of this specification.
(b) Levels of packaging and packing (see 5.1 and 5.2).
(c) Marking document (see 5.3).
(d) Registration required (see 3.1).
(e) Labeling required (see 5.4).
(f) Packing grade (level A only) required (see 5.2.1).

Custodians:

Army - GL
Navy - SH
Air Force - 68

Preparing activity:

GSA-FSS

Review activities:

Army - MD
Navy - YD, MS

Civil agency coordinating activities:

GSA-FSS
HEW-FDA, NIH, PHS
TVA
VA-DMS

User activities:

Army - ME
Navy - MC

Military coordinating activity:

Army - GL

9

Orders for this publication are to be placed with General Services Administration, acting as an agent for the Superintendent of Documents. See Section 2 of this specification to obtain extra copies and ohter documents referenced herein. Price 35 cents each.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T11:16Z
Check the source to verify that this is the current version before use.